

BM4152 Biosignal Processing Assignment 2

- This is an individual assignment.
 - Use the data files (.npz) that are attached as instructed throughout the assignment.
 - Submission guidelines:
 - Report: Upload a PDF to Moodle. Include observations and discussions with relevant plots to support your answers.
 - Python scripts / notebooks: Upload a single ZIP file with all .py files (or a single .ipynb notebook) to Moodle. Name each script/section according to the question number. Include comments for readability.
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Introduction

This assignment involves implementing the Wiener filter, as well as adaptive noise cancellers based on the LMS and RLS algorithms.

1 Wiener Filter

1.1 Data Construction

Load the `idealECG.npz` file using `scipy.io.loadmat`. Construct the following signals.

- Ideal signal $y_i(n)$: `idealECG.npz`. Sampling frequency $f_s = 500$ Hz.
- Input signal $x(n) = y_i(n) + \eta(n)$
- Where $\eta(n) = \eta_{wg}(n) + \eta_{50}(n)$
- $\eta_{wg}(n)$: white Gaussian noise such that SNR is 10 dB with respect to $y_i(n)$
- $\eta_{50}(n) = 0.2 \sin(2\pi \cdot 50 \cdot n)$

1.2 Discrete time-domain implementation of the Wiener filter

Consider the two cases given in Part 1 and Part 2 below. For each case, do the following:

- a) Calculate the optimum weight vector $\mathbf{w}_0 = (\Phi_Y + \Phi_N)^{-1} \Phi_{Yy}$ for an arbitrary filter order.
- b) Find the optimum filter order and its coefficients. Plot the magnitude response.
- c) Obtain the filtered signal $\hat{y}(n)$ using the filter obtained above.
- d) Plot the spectra of $y_i(n)$, $\eta(n)$, $x(n)$, $\hat{y}(n)$ on the same plot.
- e) Interpret the spectra and the magnitude response of the filter, given the signals and parameters in Part 1 and Part 2 below.

Part 1:

- Desired signal: a single beat of $y_i(n)$.
- Noise signal: extract the signal segment from the T wave of any chosen ECG beat to the P wave of the next ECG beat (isoelectric segment) of $x(n)$.

Part 2:

- Desired signal: Construct a linear model with the same length and a comparable morphology to that of a single ECG beat of the ideal signal $y_i(n)$. See Figure 1 for an example.
- Noise signal: Extract the signal segment from the T wave of any chosen ECG beat to the P wave of the next ECG beat (isoelectric segment) of $x(n)$.

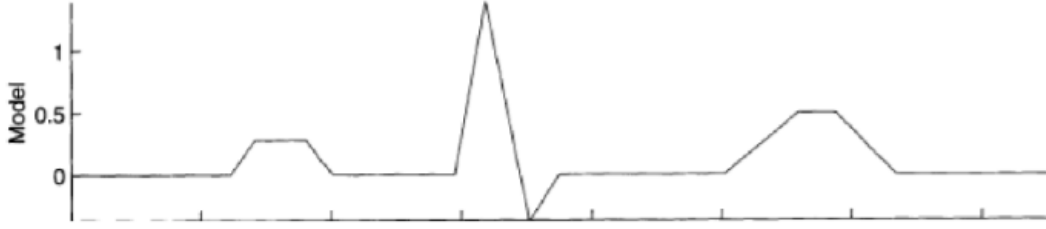


Figure 1: Piecewise-linear ECG beat model used as the desired signal in Part 2.

1.3 Frequency domain implementation of the Wiener filter

Consider the equation to calculate the weights of the Wiener filter in the frequency domain given by

$$W(f) = \frac{S_{YY}(f)}{S_{YY}(f) + S_{NN}(f)}$$

where $S_{ZZ}(f)$ is the PSD of the signal $z(n)$. Alternatively, $S_{ZZ}(f)$ is the power of the Fourier transform ($\mathcal{F}\{\cdot\}$) of the template $z(n)$. Therefore, $S_{ZZ}(f) = |\mathcal{F}\{z(n)\}|^2$.

- Implement the frequency-domain version of the Wiener filter $S_{\hat{Y}\hat{Y}}(f) = W(f) S_{XX}(f)$ for the two cases given in Part 1 and Part 2 above.
- Compare the filtered signal from Part 1 and Part 2 with the aid of a plot and the mean squared error with respect to the ideal signal $y_i(n)$.

1.4 Effect of non-stationary noise on Wiener filtering

Now consider the case where the 50 Hz noise is changed to 100 Hz halfway through the signal duration (T).

$$\eta(n) = \begin{cases} \eta_{wg}(n) + \eta_{50}(n), & 0 \leq n < T/2 \\ \eta_{wg}(n) + \eta_{100}(n), & T/2 \leq n < T \end{cases}$$

- Apply one of the Wiener filters derived in Sections 1.1 or 1.2 on the new $x(n)$. Do **NOT** recalculate the weights.
- Plot the filtered time-domain signal and interpret the result.

2 Adaptive Filtering

2.1 Data Construction

Construct the following signals.

- Sampling frequency $f_s = 500$ Hz
- Number of samples $N = 5000$
- $y_i(n)$: sawtooth waveform with a width of 0.5
- $\eta(n)$: non-stationary noise, same as in Section 1.3
- $r(n) = a(\eta_{wg}(n) + \sin(2\pi 50n + \phi_1) + \sin(2\pi 100n + \phi_2))$
where a, ϕ_1, ϕ_2 are arbitrary constants.

2.2 LMS method

- a) Implement the LMS method as per the equation

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu e(n)\mathbf{R}^T(n)$$

- b) Plot the signals $y_i(n)$, $x(n)$, $e(n) = \hat{y}(n)$, and the absolute error $|y_i(n) - \hat{y}(n)|$. Your plots should be similar to Figure 2 below.
- c) Explore the rate of adaptation by varying the rate of convergence μ and the order of the adaptive filter $(M-1)$. To quantify you may calculate the mean squared error with respect to the desired signal.

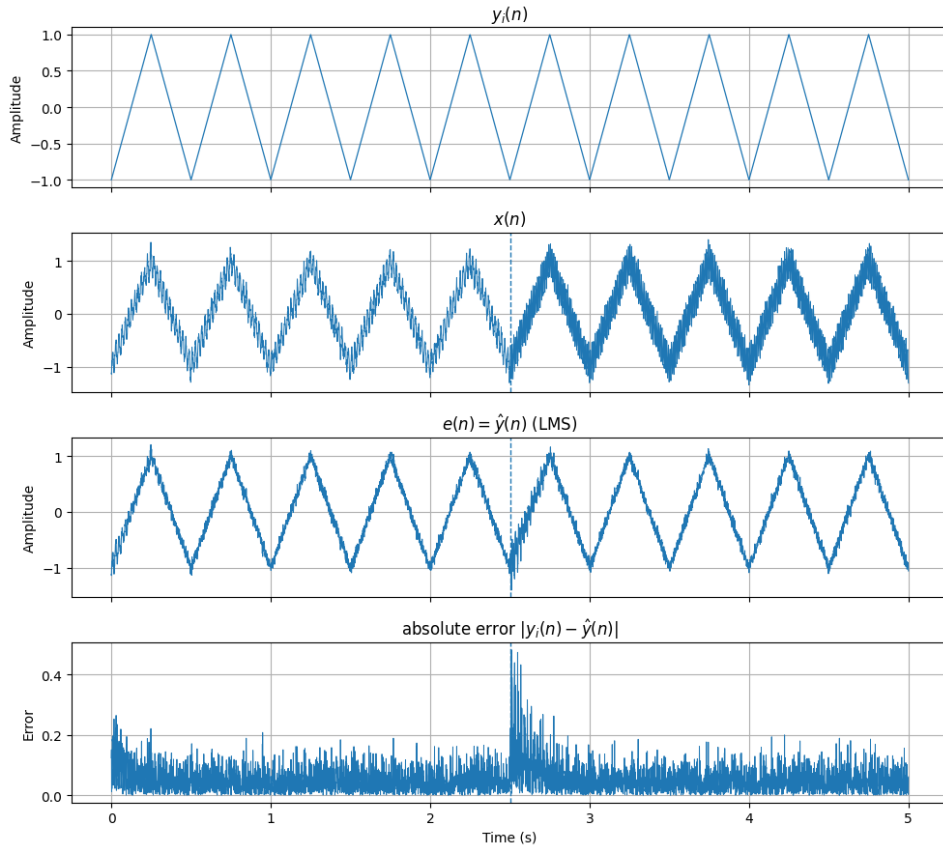


Figure 2: Reference LMS output for the sawtooth test signal: $y_i(n)$, $x(n)$, $e(n) = \hat{y}(n)$, and the absolute error.

2.3 RLS method

a) Implement the RLS method as per the equations:

$$\begin{aligned}\mathbf{k}(n) &= \frac{\lambda^{-1}\mathbf{P}(n-1)\mathbf{r}(n)}{1 + \lambda^{-1}\mathbf{r}^T(n)\mathbf{P}(n-1)\mathbf{r}(n)} \\ \mathbf{P}(n) &= \lambda^{-1}\mathbf{P}(n-1) - \lambda^{-1}\mathbf{k}(n)\mathbf{r}^T(n)\mathbf{P}(n-1) \\ \alpha(n) &= x(n) - \mathbf{w}^T(n-1)\mathbf{r}(n) \\ \mathbf{w}(n) &= \mathbf{w}(n-1) + \mathbf{k}(n)\alpha(n) \\ \alpha(n) &= \hat{s}(n) = x(n) - \mathbf{w}^T(n)\mathbf{r}(n)\end{aligned}$$

Initializations:

$$\begin{aligned}\mathbf{P}(0) &= \delta^{-1}\mathbf{I} \quad \text{where } \delta \text{ is a small constant and } \mathbf{I} \text{ is the identity matrix} \\ \mathbf{w}(0) &= \mathbf{0} \text{ or a random weight vector if necessary}\end{aligned}$$

- b) Compare the performance of LMS and RLS algorithms using a plot similar to Figure 2.
- c) Explore the rate of adaptation by varying the forgetting factor λ and the order of the filter $(M-1)$. To quantify you may calculate the mean squared error with respect to the desired signal.
- d) Now test LMS and RLS algorithms using $y_i(n)$ as `idealECG.mat` (noise $\eta(n)$ remains the same as before).